Sahaj Bhadja

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Summary

Full-stack software engineer with 2+ years building consumer-facing React applications, scalable REST services, PostgreSQL-backed systems, data-ingestion pipelines, and AI-powered search workflows using Java, Python, TypeScript, cloud infrastructure, and automated testing.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Data Retrieval, Operating Systems

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant PostgreSQL, Azure SQL, and Blob Storage services for 500+ users, optimizing schemas, indexed queries, and access patterns to reduce high-traffic dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI ingestion workflows for 5,000+ monthly jobs using validation, reconciliation, and failure isolation, achieving 99%+ production reliability while improving operational monitoring and on-call debugging visibility.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python REST APIs and instructor-facing React workflows handling 10,000+ daily requests, converting complex grading requirements into intuitive upload, review, history, feedback, and reporting experiences.
- Designed concurrent AWS RDS-backed ingestion for 1,000+ files per batch using multithreading and fault-tolerant execution, reducing turnaround by 60% while maintaining stable service performance under peak workloads.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release reliability across Windows, Linux, and macOS while collaborating in Agile code reviews.
- Optimized REST API and ETL pipelines using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving large-scale data-processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a MERN application with REST APIs, WebSockets, and eight modular services, supporting 2M+ synchronized records while translating requirements into responsive customer, inventory, billing, and notification workflows.
- Automated Docker-based builds, testing, and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, frontend-backend integration, and defect resolution across production releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs for a cross-platform booking application supporting search, scheduling, reservations, pricing, payments, and status updates across 50,000+ records with synchronized Firebase-backed user workflows.
- Implemented webhook-based payment synchronization and operational troubleshooting, reducing manual payment handling by 70% while improving transaction visibility, error recovery, and responsiveness across mobile and web experiences.

Skills

Full Stack: React, TypeScript, JavaScript, Java, Python, SQL, HTML5, CSS3, Spring, Node.js, FastAPI, Flask, REST APIs, JSON, WebSockets, microservices, responsive UI, object-oriented programming; **Search & Data:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Firebase, AWS RDS, data modeling, indexing, query tuning, ETL, data ingestion, information retrieval, semantic search, vector search, recommendation systems, NLP, LLM orchestration, analytics; **Reliability:** AWS, Azure, GCP, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, unit testing, integration testing, debugging, monitoring, fault tolerance, performance optimization, Agile/Scrum, code review, AI-assisted development

Projects

Big-Brain Live | *Python, LLaMA-2, FAISS, Pinecone, PostgreSQL*
- Built an AI search platform for 200+ users with secure ingestion, vector indexing, semantic retrieval, and collaborative workflows, maintaining 95% query accuracy across 50,000+ searchable documents.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker, REST APIs*
- Built a Java and TypeScript database workspace supporting 10+ systems, accelerating SQL authoring, schema exploration, validation, and cross-platform data management through a unified search-oriented interface.

PiXMod Live | *Next.js, TypeScript, CLIP, MongoDB, Transformers*
- Built an AI-powered visual search application using CLIP embeddings and MongoDB, enabling semantic retrieval across 100,000+ indexed images through responsive Next.js workflows and scalable metadata pipelines.